Seat No._____

GANPAT UNIVERSITY
B.Sc. (CA & IT) SEM-III EXAMINATION
NOV- DEC 2019
U13A1DS DATA STRUCTURE

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. What is data structure? Explain the types of it.
2. Write down algorithm for Deque insert and delete operations.
3. Explain different tree traversal method with example
4. Give the difference between SMA and DMA.
5. Write down the algorithms for all stack operations.
6. Explain binary search with example.
7. Explain selection sort with suitable example.

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Write down algorithm for circular queue insert and delete operations.
2. Write down algorithm for Merging of two simple link lists.
3. Write down algorithm to insert a node at the beginning of the single link list.

3. Answer the following: (Any Two) **(12)**

1. Write down algorithm to insert the node in tree.
2. Explain Threaded tree.
3. What is sequential search? Explain with example and algorithm.

4. Answer the following: (Any One) **(06)**

1. Explain bubble sort with example.
2. Sort following data using Insertion sort.

   44 33 11 55 77 90 40 60

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. (CA & IT) SEM-I EXAMINATION
NOV- DEC 2019
U11A2OAT:OFFICE AUTOMATION TOOLS

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. **Answer the following: (Any Five)** **(30)**

1. What are the differences between CUI and GUI?
2. Describe DIR command with all attributes.
3. What is mail merge? Write down steps for mail merge.
4. Discuss importance of Freeze Panes and Drawing Tools in Excel.
5. What is Excel? Define features of Excel.
6. Discuss features of PowerPoint.
7. Discuss Data types of Microsoft Access.

SECTION-II

2. **Answer the following: (Any Two)** **(12)**

1. Describe all the steps for creating Batch file.
2. What are the difference between internal Vs. external Command.
3. Explain Microsoft word's features in detail.

3. **Answer the following: (Any Two)** **(12)**

1. Define terms: Goal Seek, Pivot Table.
2. Discuss Mathematical functions of Excel.
3. Discuss the Find and Replace Text function of Excel.

4. **Answer the following: (Any One)** **(06)**

1. Describe steps to creating Database in Microsoft Access.
2. Discuss different elements of Microsoft Access.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. (CA & IT) SEM-III EXAMINATION
NOV- DEC 2019
U13A4SAD SYSTEM ANALYSIS AND DESIGN

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. **Answer the following: (Any Five)** **(30)**

1. Define System and Characteristic of a System.
2. Explain Need for System Development.
3. List out various Fact finding techniques and describe any one in brief.
4. Explain various level of Testing in detail.
5. Explain testing process in brief.
6. Explain Decision Table in detail.
7. Explain output design in brief.

SECTION-II

2. **Answer the following: (Any Two)** **(12)**

1. What is DFD? Describe Symbols Used in DFD with Example.
2. Write down Difference between Physical and Logical DFD.
3. What is System Analyst? Explain Role of System Analyst.

3. **Answer the following: (Any Two)** **(12)**

1. Explain various types of codes in brief.
2. Explain data validation in detail.
3. Explain white box and black box testing in detail.

4. **Answer the following: (Any One)** **(06)**

1. Draw Case Study for Hospital Management System.
2. Draw case study for Railway Reservation system.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. (CA & IT) SEM-III EXAMINATION
NOV- DEC 2019
U13A3DMS: DATABASE MANAGEMENT SYSTEM

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. Explain Dr. E.F. Codd's rules for RDBMS
2. Short note on Network Data Model
3. Discuss 3 NF with an example.
4. Explain various data types in Oracle.
5. Discuss aggregate functions.
6. Short note on Set Operators.
7. Discuss GRANT and REVOKE statement in detail.

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Explain Advantages of DBMS.
2. E-R Diagram in details.
3. Write short note on Join Dependencies.

3. Answer the following: (Any Two) **(12)**

1. Explain Primary key and Foreign key with an example
2. Discuss DDL commands
3. Explain various types of joins.

4. Answer the following: (Any One) **(06)**

1. Explain Index in detail.
2. Discuss Sequence syntax with an example.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. (CA & IT) SEM-III EXAMINATION
NOV- DEC 2019
U13A2OCP: OBJECT ORIENTED CONCEPTS AND PROGRAMMING

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. Explain Features of OOP.
2. How StringBuffer class is different from String class? Discuss any two methods that is not available in String class.
3. What is Throwable? Explain try, catch finally, throw and throws.
4. What is method overloading? Explain with example
5. Explain with example Vector class.
6. What is the use of Lambda expression? In Which Version of jdk Oracle introduced this features? Explain in brief block lambda expression.
7. Explain phaser in detail.

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Why java is platform independent language? Explain JDK,JRE, JRE, JIT
2. Which are the ways to implements thread in java? Explain with example.
3. Explain access specifier in java.

3. Answer the following: (Any Two) **(12)**

1. How to create object of Calendar class? List out any six constants of Calendar.
2. Explain method reference in lambda expression.
3. Differentiate HashTable and Vector. Explain HashTable in detail

4. Answer the following: (Any One) **(06)**

1. Discuss fork/join framework
2. Explain executer in detail.

END OF PAPER